

Effectiveness of Visual Acuity Tests in VR vs the Real-World

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ABSTRACT

Virtual Reality (VR) devices have opened a new dimension of merging technology and healthcare in an immersive and exciting way to test eye vision. Visual acuity is a person's capacity to perceive small details. An optometrist or ophthalmologist determines a visual acuity score following a vision examination. In this work, we explored how recent VR devices could be utilized to conduct visual acuity tests. We used two Snellen charts to examine eye vision in VR, similar to testing in a doctor's chamber. We found that VR could be utilized to conduct preliminary eye vision tests.

Keywords: Eye Vision Test; Virtual Reality; User Experiment.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Virtual reality; Applied computing—Life and medical sciences—Consumer health;

1 INTRODUCTION

Eye vision is an important area of research due to its significance in our daily lives and overall well-being. The goal of VR eye vision research is to better understand the complexities of vision and uncover factors that can impact visual health. Recently Wang et al. [4] found that reading may be a tedious and frustrating process for those with limited vision and they require more advanced technologies to improve the reading experience. Shekar et al. [3] developed a VR-based vision test for all age groups but they had issues in accurately scaling real-world distances into a corresponding virtual distance scale. Another study [2] investigated how much the visual acuity is reduced in VR due to the low resolution of the VR display. In this study, we wanted to see if a recent VR device with better resolution could be utilized for a visual acuity test to get a preliminary idea before an eye doctor's visit.

Visual acuity is evaluated [1] using terms like oculus uterque (OU) for both eyes, oculus sinister (OS) for the left eye, and oculus dexter (OD) for the right eye. The results of a visual acuity test depend entirely on the distance that a "normal" individual can see objects at, which is 20 feet (6 meters). If a patient's visual acuity is recorded as 20/20 or 6/6, this indicates that they can see what a normal person would typically see at a distance of 20 feet. If a person's vision is described as 20/40 or 6/12, they can see an image or item at a distance of 20 feet that a typical person would only be able to recognize at a distance of 40 feet or 6 meters instead of 12 meters, respectively. A professional eye doctor or optometrist measures eyesight and gives a score based on how well a person can read the chart. We wanted to see if that can be done with VR devices as an alternative solution to visiting doctors. Thus, our initial research question was to what extent could we conduct eye vision tests in VR?

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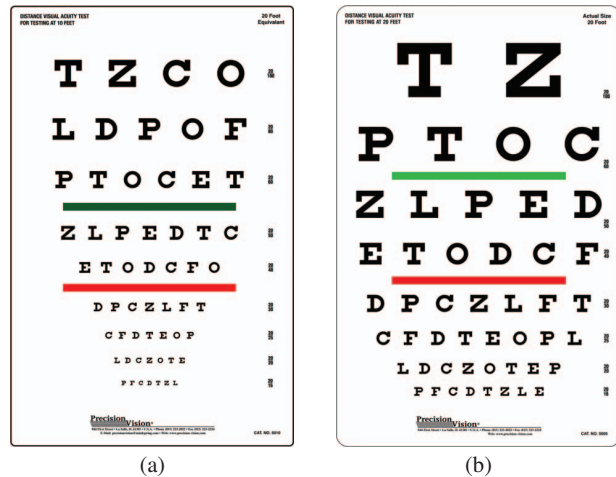


Figure 1: Eye vision test with Snellen Charts: (a) test chart for 10 feet distance, and (b) test chart for 20 feet distance.

2 USER EXPERIMENT

Two charts (see Figure 1) were used in our experiment; one was for a distance of 20 feet from the participant, while the other was for a distance of 10 feet. We created a VR environment where a vision test chart is shown in VR (see Figure 2). For each chart, we had three conditions: 1) a real chart displayed in a room with natural lighting conditions, 2) a real chart displayed in a VR scene in JPG format, and 3) a real chart in a VR scene in SVG format. In the first condition, similar to a doctor's office, a printed chart was utilized. The size of chart one was 14 by 9 inches, and the size of chart two was 11 by 7 inches (see figure 1). In the second condition, an image (JPG format) chart was used in the VR environment. The original JPG chart was adjusted in size to match the real-world resolution. The VR headset resolution was taken into consideration, accounting for its lower pixels per inch (PPI) compared to the real-world resolution. The third condition used an SVG image in the VR environment. The SVG format allowed for a reasonable resolution without the need for upscaling the image. This approach aimed to maintain a more realistic chart size in VR compared to the real-world chart while preserving the same ratio size as the original chart.

For each condition, we collected participants' scores for individual eyes and post-questionnaire feedback (see Table 1) about the overall experience. The scores were calculated based on when the participant missed more than 2 letters in a row. They use one eye at a time and cover the other eye with a piece of paper. Participants read the letters from the top left and press the button to match what they see in the charts in VR. If they press the button incorrectly more than 2 times, the score is recorded for that eye. Each row has a different score associated with it, indicated on the right side of each row in the chart. The participants may have low visual acuity if they are unable to read below the Green line (see Figure 1) or any line below the 20/60 mark on the eye chart. This tool will measure how well each eye sees, and the results of these tests are portrayed as a fraction, with 20/20 being the standard for normal distance and reading vision.

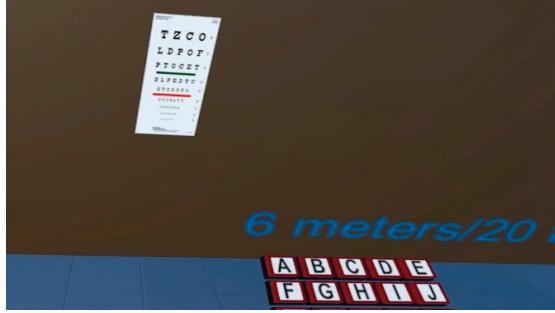


Figure 2: Eye vision test in VR environment at 20 feet distance.

Table 1: Post-questionnaire questions. Participants answered Q1-Q2 on a 7-point Likert-like items.

Post-questionnaire Questions	
Q1	To what extent did the VR eye vision test similar to the real-world eye vision test?
Q2	Do you think VR can perform a basic eye vision test before you see the doctor?
Q3	In case of 10 feet distance, which VR chart is more close to a real chart? a) Scaling chart b) Regular chart
Q4	In case of 20 feet distance, which VR chart is more close to a real chart? a) Scaling chart b) Regular chart
Q5	Overall comment for VR eye vision test experience.

For this experiment, we used a Meta Quest Pro VR headset, a desktop computer (Core i9 11900F, Microsoft Windows 10 Pro, NVIDIA GeForce RTX 3080 Ti, 64 GB RAM), and Unity 3D v2022.3.16f1 software to implement the VR tasks. Nine participants (7 males and 2 females, age range: 20 to 31), attended our pilot study, which lasted for 15 to 25 minutes. Three participants wear eyeglasses prescribed by doctors but they did not use glasses in this experiment. For each participant, we calibrated the VR headset and ensured a proper fit, including adjusting the interpupillary distance (IPD) so that they could see the charts clearly.

3 PRELIMINARY STUDY RESULTS

Based on the post-questionnaire Q1 ratings, most participants found the real-world tests and VR tests very similar (mean rating of 5.11 and $\sigma = 1.36$). Based on Q2 ratings, most participants agreed (mean rating of 5.11 and $\sigma = 1.05$) that a VR-based vision test could suffice as a preliminary test before they decide to visit an eye doctor. For the 10-foot distance (Q3), five participants preferred the JPG format chart, three preferred the SVG format chart and one could not decide to vote. At a 20-foot distance (Q4), six preferred the JPG format chart, two preferred the SVG format and one did not vote. For Q5, 5 of 9 participants commented that this is a good tool that can be applied to test eye vision in a VR environment. Based on the participants' feedback, the results of these studies showed that the VR vision test is similar to the traditional vision test.

According to the visual acuity results (see Figure 3 and 4), we see that most of the participants had similar scores at 10 and 20-foot distances for all conditions. However, there are small differences for some participants. We see that at 10 feet distance, the score difference for each condition is only one level either up or down except for participant 2 (3-level difference). On the other hand, at 20 feet distance, the score difference between the real-world condition and VR JPG condition is only one level either up or down except for Participants 2 and 5 (3 level difference). When comparing the real-world condition with the VR SVG condition, the difference is more than 3 levels for three participants (1, 2, and 5). Participants 2, 5, and 9 used eyeglasses regularly but they did not use glasses to

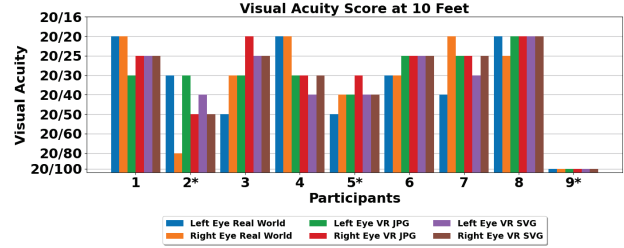


Figure 3: Visual Acuity Scores at 10 feet. The * indicates participants who use eyeglasses but did not use them for this experiment.

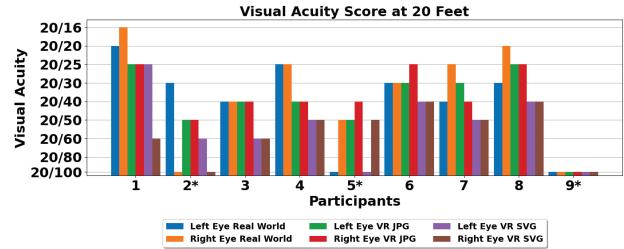


Figure 4: Visual Acuity Scores at 20 feet. The * indicates participants who use eyeglasses but did not use them for this experiment.

read the chart in this experiment. Our VR tests were able to find that their vision is not normal. Thus, we believe that people with limited vision can utilize this tool for preliminary tests to get an initial idea. However, future testing with statistical analysis is necessary with more participants.

4 CONCLUSION AND FUTURE WORK

Our pilot study results indicate that VR-based visual acuity tests are a viable option as a preliminary test before visiting an eye doctor. Although, these tests may not be as accurate as physical Snellen charts, for most of the participants this VR tool would help them get a rough visual acuity score. We assumed that the resolution of the VR device used (Meta Quest Pro) and the effect of the lenses made the chart clearer and therefore easy to read. However, some concerns may arise due to device calibration, scaling issues with the image, and Inter-pupillary Distance (the distance between the eyes). Fitting the device correctly on the head could reduce some of these effects. Further tests would be needed to find out if better devices or other factors could allow VR to have the potential to be a great tool for vision tests. Moreover, integrating eye tracking and investigating the pattern of low-vision people's gaze behavior could provide better ideas for early detection of vision problems.

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REFERENCES

- [1] D. Caltrider, A. Gupta, and K. Tripathy. Evaluation of visual acuity. 2020.
- [2] L. Panfili. Effects of vr-displays on visual acuity. 2019.
- [3] S. Shekar, P. R. Pesaladinne, S. A. Karre, and Y. R. Reddy. Vreye: Exploring human visual acuity test using virtual reality. In *Virtual, Augmented and Mixed Reality. Industrial and Everyday Life Applications: 12th International Conference, VAMR 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Copenhagen, Denmark, July 19–24, 2020, Proceedings, Part II* 22, pages 415–429. Springer, 2020.
- [4] R. Wang, L. Zeng, X. Zhang, S. Mondal, and Y. Zhao. Understanding how low vision people read using eye tracking. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pages 1–17, 2023.